

ORIGINAL ARTICLE

Ultrasensitive and affordable assay for early detection of primary liver cancer using plasma cell-free DNA fragmentomics

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Abstract

Background and Aims: Early detection of primary liver cancer (PLC), including HCC, intrahepatic cholangiocarcinoma (ICC), and combined HCC-ICC (cHCC-ICC), is essential for patients' survival. This study aims to develop an accurate and affordable method for PLC early detection and differentiating ICC from HCC using plasma cell-free DNA (cfDNA) fragmentomic profiles.

Approach and Results: Whole-genome sequencings (WGS) were performed using plasma cfDNA samples from 192 patients with PLC (159 HCC, 26 ICC, 7 cHCC-ICC) and 170 noncancer controls (including 53 liver cirrhosis [LC] or HBV-positive) enrolled in the training cohort. An ensemble stacked model for PLC detection was constructed using the training cohort. The model performance was assessed in an independent test cohort (189 patients with PLC [157 HCC, 26 ICC, 6 cHCC-ICC], 164 noncancer controls [including 51 LC/HBV]). Our model showed excellent performance for cancer detection in the test cohort (AUC: 0.995, 96.8% sensitivity at 98.8% specificity). It showed excellent sensitivities in detecting early-stage PLC (I: 95.9%, II: 97.9%), small tumors (≤ 3 cm: 98.2%), and HCC (96.2%) or ICC (100%). The AUC for distinguishing PLC from LC/HBV reached 0.985 (96.8% specificity at 96.1%

Abbreviations: AFP, alpha-fetoprotein; AJCC, American Joint Committee on Cancer; BCLC, Barcelona Clinic Liver Cancer; cfDNA, cell-free DNA; cHCC-ICC, combined HCC-ICC; CNLC, China Liver Cancer; DELFI, DNA evaluation of fragments for early interception; FN, false-negative; FP, false-positive; FSD, Fragment Size Distribution; FSR, Fragment Size Ratio; GBM, gradient boosting machine; GLM, generalized linear model; ICC, intrahepatic cholangiocarcinoma; LC, liver cirrhosis; PLC, primary liver cancer; ROC, receiver operating characteristic; TN, true-negative; TP, true-positive; WGS, whole-genome sequencing.

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specificity). Promisingly, our model maintained consistent performances during the downsampling process, even using 1X coverage data (AUC: 0.994, 93.7% sensitivity at 98.8% specificity). A separate model showed potential for distinguishing ICC from HCC (AUC: 0.776).

Conclusions: Our model, outperforming previous reports at a lower cost by solely using low-coverage WGS data, exhibits excellent clinical potential for ultrasensitive and affordable detection of PLC and its subtypes.

INTRODUCTION

Primary liver cancer (PLC) is a leading cause of cancer mortality worldwide, with a total of over 840,000 new cases identified and 780,000 related deaths annually.^[1] PLC exhibits a poor prognosis with an estimated 5-year net survival of 19%, whereas its incidence rates are high and still rising in many areas of the world.^[2,3] There are two major histological types of PLC: HCC (~75% of all liver cancers) and intrahepatic cholangiocarcinoma (ICC; ~12%–15% of all liver cancers).^[3] Differentiating HCC and ICC is also essential in clinical practice because the treatment for these two conditions differs significantly.^[4] For disease management, early detection of PLC is essential for curative treatment and patient survival.^[5,6] In the past few decades, novel imaging techniques and serologic markers have been introduced to increase the diagnostic capacity. However, many of the techniques are not optimal for PLC screening, and their applications are restricted by different factors such as lack of sufficient sensitivity or specificity and radiation exposure. For example, the identification of microscopic PLC lesions, especially on the background of cirrhosis, remains challenging for the radiological approaches.^[7]

Recently, the study of cell-free DNA (cfDNA) has provided a noninvasive approach for the diagnosis of solid malignancies such as PLC in liquid biopsies. cfDNAs represent extracellular DNA fragments released from cell apoptosis and necrosis into human body fluids like plasma,^[8] therefore carrying the genetic and epigenetic information from the cell and tissue of origin.^[9] Specific somatic mutations previously localized in the primary tumor tissue have been used as biomarkers to distinguish tumor and nontumorous cfDNA,^[10] but the usage of the somatic mutations is limited by their variability and low fractional concentration in plasma.^[11] Alternatively, different molecular signatures present on cfDNA, including DNA methylation patterns and tumor-associated copy number aberrations, are related to cancer pathobiology and have been shown as feasible biomarkers in HCC.^[12–15] Several groups have also employed the fragmentomic features of cfDNA,^[16] such as genome-wide profiling of cfDNA fragmentation size

and end motif, to detect cancer at an early stage.^[17,18] In 2019, Cristiano et al. introduced an approach named DELFI (DNA Evaluation of Fragments for early Interception), which focused on fragmentation size ratio within different bins in the genome.^[18] The DELFI approach, which targeted a multicancer patients cohort, was able to accurately detect 88% of the 26 patients with ICC in their cohort at the specificity of 95%.^[18] They later reported the utilization of fragmentomics in a lung cancer cohort, achieving an overall AUC of 0.90.^[19]

Despite considerable progress in biomarker discovery, the application of new cfDNA features is restricted due to a lack of sensitivity, especially for detecting PLC at the early stage. Xu et al. demonstrated a sensitivity of 83% at the 90% specificity for HCC detection using DNA methylation profiling, and the detection performance further decreased for patients with Stage I and II disease.^[12] Cai et al. used genome-wide 5-hydroxymethylcytosine profiling and reached a sensitivity of 83% at the specificity of 76% (AUC = 0.88) for early-stage HCC. Using the 4-nucleotide sequence of cfDNA 5' end, Jiang et al. distinguished HCC from non-HCC controls, but the detection AUC is as low as 0.86.^[13,17]

Lately, a model integrating several cfDNA molecular features has shed some light on leveraging HCC detection performance, exhibiting 95% detection sensitivity at the specificity of 97%.^[20] However, this model requires two different NGS protocols, which drastically increases the cost and time for this method. Currently, a fast, affordable, and accurate model is still needed for PLC early detection.

In this prospective study, we constructed a fragmentomics-based machine learning model for PLC detection using low-coverage whole-genome sequencing (WGS) data to overcome the aforementioned setbacks by providing an ultrasensitive and affordable method. Our method consistently exhibited improved prediction performance of detecting PLC, presenting higher specificity and sensitivity than the previous reports, even when using extreme shallow WGS data as low as 1X coverage. Furthermore, our method has illustrated the potential of using cfDNA profiling to differentiate HCC and ICC.

PATIENTS AND METHODS

Study design and patient enrollment

The cohort sizes were determined using the R package MKmisc (version 1.8), designed for sample size calculation in binary classification tests^[21,22] and commonly used in clinical trials.^[23] Based on the calculation, ~193 patients with PLC and ~167 noncancer controls (patients with liver cirrhosis [LC]/HBV and healthy volunteers) were sufficient for testing a desired 97% sensitivity at 98% specificity (margin of error = 5%, significance level = 0.05, power of test = 0.9). The participants enrolled in this study were assigned to an independent training and an independent testing cohort at a ~1:1 ratio by the personnel who were collecting the clinical information. A total of 362 participants, including 192 patients with PLC, 53 patients with LC or HBV (LC/HBV), and 117 healthy volunteers, were enrolled in a training cohort between June 2019 and December 2020, which was used to construct a machine learning model. The independent test cohort enrolled 354 participants (189 PLC, 51 LC/HBV, 114 healthy) for evaluating model performances. The team involved in cohort design was blinded to the sequencing, feature extraction, model construction, and validation process. The participants were enrolled from Liver Cancer Institute, Zhongshan Hospital.

All study protocols were approved by the ethics committee of Zhongshan Hospital Fudan University (Approval No. B2019-121R) and in accordance with the principles of the 1975 Declaration of Helsinki. Written informed consent was provided by all patients and no donor organs were obtained from executed prisoners or other institutionalized persons. Approximately 10 ml peripheral blood samples were collected from each participant at the time of diagnosis. The blood samples were shipped overnight at -80°C to the central laboratory of a clinical testing center (Nanjing Geneseeq Technology Inc., China) for plasma preparation and cfDNA extraction within 48 h of collection.

WGS and bioinformatics processing

Available peripheral blood samples from participants in this study underwent cfDNA extraction (~2.5 h) followed by WGS. The plasma samples were centrifuged at 16,000× for 10 min to remove cell debris, followed by cfDNA extraction using QIAamp Circulating Nucleic Acid Kit (Qiagen) according to the manufacturer's protocols. The cfDNA concentrations in these plasma samples were examined with Qubit dsDNA HS Assay Kit (Thermo Fisher Scientific) as per manufacturer guidelines. WGS libraries were constructed (~6 h) using the

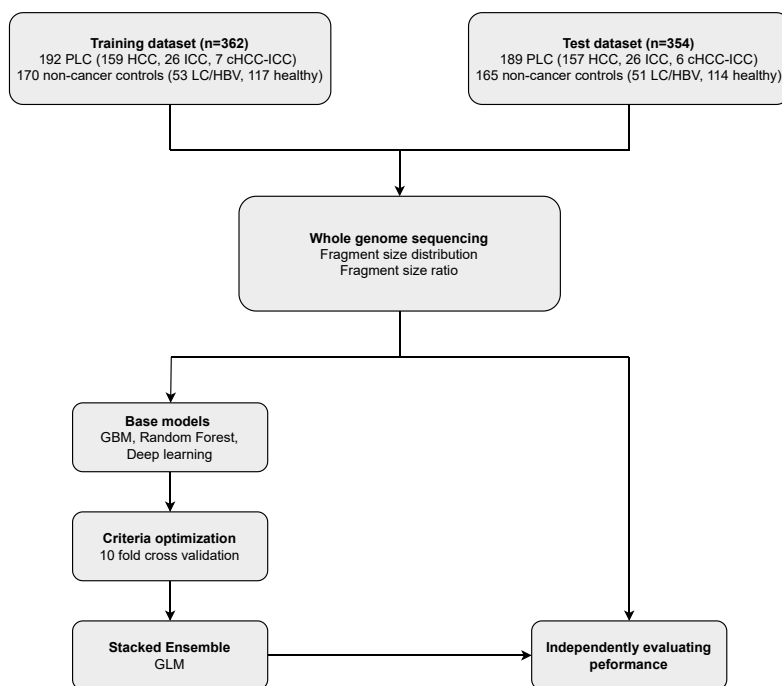
KAPA Hyper Prep Kit (KAPA Biosystems) according to the manufacturer's protocol. In brief, 5–10 ng of cfDNA per sample was subjected to end-repairing, A-tailing, and ligation with adapters sequentially. The Hamilton Microlab STAR automated liquid handling platform (Hamilton Company) was used for the automated pipeline. The libraries were quantified (~3 h) with the KAPA SYBR FAST qPCR Master Mix (KAPA Biosystems) and underwent paired-end sequencing on NovaSeq platforms (Illumina) according to the manufacturer's instructions (~40 h).

Raw sequencing data were trimmed by Trimmomatic as part of the quality control protocol.^[24] The qualified reads were then mapped onto the human reference genome (GRCh37/UCSC hg19) using the sequence aligner BWA^[25] after PCR duplicates removal by Picard toolkit (<http://broadinstitute.github.io/picard/>) (~4 h). The final coverage depths for these samples ranged from 4.45X to 24.04X (median coverage 9.935X). To eliminate the potential impact on the predictive power by the different coverages among the WGS data, we down-sampled the coverages to a unified 4X, ensuring the inclusion of all samples.

Two fragment size profiles, Fragment Size Ratio (FSR) and Fragment Size Distribution (FSD), were adapted from the DELFI fragment size profile and optimized in order to improve its predicting ability,^[18] as shown in Figure 1. The FSR feature examined the ratios of different size fragments across the human genome, as cfDNA fragments could be aberrantly short and long in the plasma sample from patients with cancer.^[16] The fragment sizes were used to construct these fragmentation profiles using in-house scripts (Figure 1). The FSR, similar to the original fragmentation profile reported by DELFI,^[18] was generated using the short/intermediate/long fragments ratios except using different cutoffs: the short, intermediate, and long fragments were defined as 65–150 bp, 151–220 bp, and 221–400 bp, according to the overall fragment lengths profile in our cohorts. The new cutoff allowed the inclusion of larger fragment size regions compared with the reported 100–150 bp/151–220 bp, which can, in theory, improve model performances. The ratios of the short/intermediate/long fragments for each sample were examined in 5 Mb bins, resulting in a total of 1716 (572 bins × 3) FSR features from the 572 bins genome-wide. For each sample, the standard score (Z score) for each of the 1716 FSR features was calculated by comparing the short/intermediate/long feature against their mean values. These normalized Z scores were then used by the machine learning algorithms for model construction.

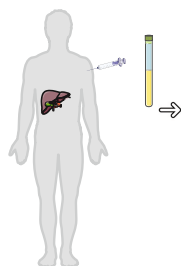
The FSD examined fragment length patterns at a higher resolution by grouping cfDNA fragments into length bins of 5 bp ranging from 65 bp and 400 bp ([400 - 65]/5 = 67 bins). The FSD was calculated using the ratio of fragments in each bin at each chromosome arm level. In total, 41 chromosome arms were examined,

(A)

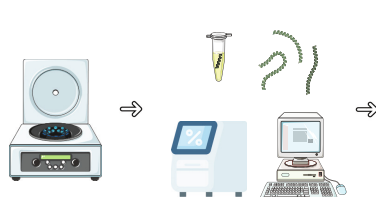


(B)

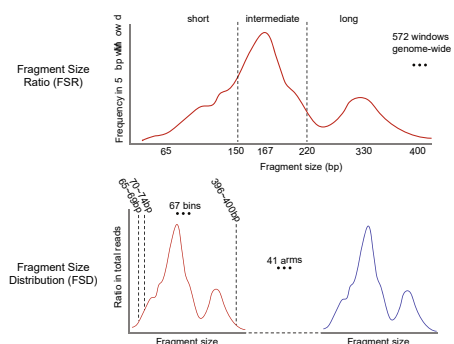
I. Plasma collection



II. Sample preparation and WGS



III. Fragmentomic feature extraction



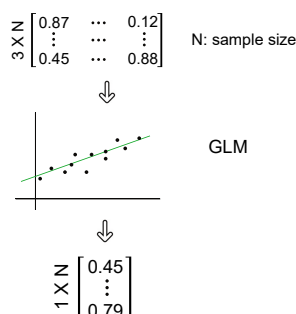
VI. Final prediction

$$2 \times N \begin{bmatrix} 0.45 & 0.23 \\ \vdots & \vdots \\ 0.79 & 0.62 \end{bmatrix} \downarrow \text{GLM}$$

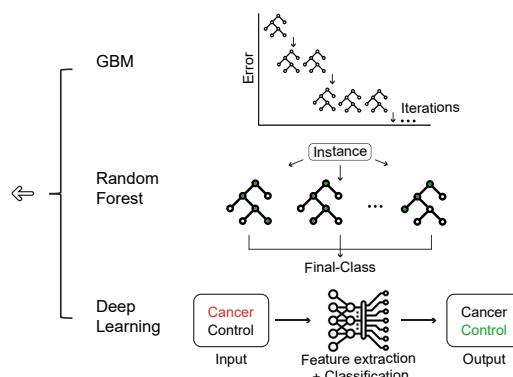
$$1 \times N \begin{bmatrix} 0.10 \\ \vdots \\ 0.67 \end{bmatrix}$$

Cancer Probability Score

V. Ensemble stacking



IV. Base models construction



generating 2747 (41 arms $\times$ 67 bins) FSD features. Similar to the FSR features, the standard scores (Z score) were calculated for each of the 67 bins across 41 chromosome arms and were subsequently used by the machine learning algorithms.

Machine learning model construction

A total of 4463 features, including the FSR (572 $\times$ 3 = 1716 features) and FSD (41 $\times$ 67 = 2747 features) profiles, were used to construct a stacked model for

FIGURE 1 Schematic representation of study and cohort design. (A) Cohort design. The training cohort ($n = 362$) included 192 patients with PLC (159 HCC, 26 ICC, 7 cHCC-ICC), 53 patients with LC/HBV, and 117 healthy controls and was used to train the ensemble stacked model. The test cohort ($n = 354$), which included 189 patients with PLC (157 HCC, 26 ICC, 6 cHCC-ICC), 51 patients with LC/HBV, and 114 healthy controls, was used to evaluate model performances. (B) Study design. Plasma samples, which were collected from patients with PLC and noncancer controls, were used to extract cfDNA, followed by WGS. The FSR feature was generated using the short/intermediate/long fragments ratios under length cutoffs 65–150 bp, 151–220 bp, and 221–400bp, according to the overall fragment lengths profile in our cohorts. For each sample, the ratio between short and long fragments was examined in 5 Mb bins, forming 572 bins genome-wide. The FSD feature was generated by grouping cfDNA fragments into length bins of 5 bp ranging from 65 to 400 bp and calculating the ratio of fragments in each bin at arm level for each chromosome. A total of 41 chromosome arms were examined. Three base models, including GBM, Random Forest, and Deep Learning, were constructed. The predictions by base models were ensembled into a large matrix, which was subsequently used by a GLM algorithm to train the final ensemble stack model

predicting PLC. The 362 participants in the training cohort, as shown in Figure 1, were categorized into cancer (192 patients with PLC) and noncancer groups (170 noncancer participants).

For each FSR and FSD feature, a generalized linear model (GLM) was constructed using predicted cancer scores from three base algorithms including gradient boosting machine (GBM), Random Forest, and Deep Learning, which could increase the model's robustness and accuracy as shown by the machine learning community.^[26] Each algorithm used PLC and noncancer control samples in the training cohort to generate cancer scores ranging from 0 to 1, whereas a higher score represented a higher probability for cancer. The cancer scores from all three algorithms were then ensembled into a matrix, which was subsequently used by a GLM algorithm to create the base model (Figure 1).

The final ensembled machine learning model was constructed by stacking both FSR and FSD base models with a second layer GLM algorithm. To optimize model performance and avoid overfitting, 10-fold cross-validation was also performed based on the training dataset. The independent test cohort, which remained untouched during the model construction, was used for evaluating the performances of the final stacked ensembled predictive model while ensuring a specificity higher than 98%.

Statistical analysis

The receiver operating characteristic (ROC) curves were constructed using the pROC package (v. 1.17.0.1). Based on true-positive (TP), true-negative (TN), false-positive (FP), and false-negative (FN) of cancer prediction, the sensitivity $[TP/(TP + FN)]$, specificity $[TN/(TN + FP)]$, positive $[TP/(TP + FP)]$ and negative $[TN/(TN + FN)]$ predictive values, accuracy $[(TP + TN)/(TP + FP + TN + FN)]$ as well as the 95% CI were calculated using the epiR package (v. 2.0.19). Propensity score matching analysis of age and sex within the test cohort was performed using the MatchIt package (4.4.2.0). All statistical analyses were performed in R (v. 3.6.3).

RESULTS

Participant characteristics in the cohorts

A total of 716 participants, including 381 patients with PLC and 335 noncancer controls, were enrolled in this study (Figure 1). As shown in Table S1 and Figure 1, the training cohort contained 362 individuals, including 192 patients with PLC (159 HCC, 26 ICC, 7 combined HCC-ICC [cHCC-ICC]) and 170 noncancer participants (53 patients with LC/HBV and 117 healthy volunteers). The test cohort ($n = 354$) included 189 patients with PLC (157 HCC, 26 ICC, 6 cHCC-ICC) and 165 noncancer controls (51 patients with LC/HBV and 114 healthy volunteers). The ratio of patients with HCC, ICC, and cHCC-ICC in both cohorts reflected their true percentages in real-world clinical settings.

For the patients with PLC enrolled in this study, the mean age was 59.2 and 59.7 years in the training and test cohorts, respectively, compared with the mean age of 48.8 and 48.6 years in the noncancer controls, as shown in Table S1. In the training and test cohorts, 20.3% and 19.6% of the patients with PLC were female, which were lower percentages compared with 54.1% and 53.9% for the noncancer controls (Table S1). Among the different types of PLC, the patients with HCC accounted for the majority, consisting of 82.8% and 83.1% of the training and test cohorts, respectively, compared with only 13.5% and 13.7% for the patients with ICC. Both cohorts mainly contained patients with early-stage PLC, with the American Joint Committee on Cancer (AJCC) TNM staging system^[27] stage IA, IB, and II adding up to more than 88% and 91% in the training and test cohorts, and the Barcelona Clinic Liver Cancer (BCLC)^[28] staging system 0/A patients accounted for 69.8% and 74.1% in the two cohorts, whereas the China Liver Cancer (CNLC) staging system^[29] early-stage (Ia and Ib) patients accounted for 70.3% and 74.1%, as shown in Table S1.

Constructing a predictive model for PLC detection

Using the FSR and FSD feature, we constructed an ensemble stacked model for predicting PLC using

down-sampled 4X coverage WGS data to eliminate potential impact by fluctuating sample coverages while ensuring the inclusion of all samples. ROC curves were constructed for the base and stacked models using the FSR and FSD profile in the test cohort to demonstrate the advantage of the ensemble stacked machine learning model approach. Furthermore, we constructed and evaluated a GBM model using the DELFI feature as reported initially by Cristiano et al.^[18]

As shown in Figure S1, the stacked model showed superior AUC (0.995; 95% CI, 0.989–1.000) than both base models using FSR (0.993; 95% CI, 0.983–1.000) and FSD (0.991; 95% CI, 0.984–0.998). The stacked model also exhibited a higher sensitivity for PLC detection (96.8%; 95% CI, 93.2%–98.8%) comparing with the base models (FSR, 93.7%; 95% CI, 89.2%–96.7%; FSD, 84.1%; 95% CI, 78.1%–89.0%) (Table S2). Also shown in Figure S1, our stacked and base model all showed higher AUC (stacked, 0.995; 95% CI, 0.989–1.000; FSR, 0.993; 95% CI, 0.983–1.000; FSD, 0.991; 95% CI, 0.984–0.998) for distinguishing PLC from noncancer controls than the original DELFI GBM model (0.989; 95% CI, 0.981–0.998) in the test cohort.

As shown in Figure 2A and Table 1, the 4X model reached an AUC of 0.995 (95% CI, 0.989–1.000) in differentiating PLC from noncancer controls in the test cohort, yielding a sensitivity of 96.8% (95% CI, 93.2%–98.8%) and a specificity of 98.8% (95% CI, 95.7%–99.9%). The differentiation power of our model for PLC versus healthy (AUC = 0.999; 95% CI, 0.997–1.000) was slightly better than for PLC versus LC/HBV (AUC = 0.985; 95% CI, 0.968–1.000), as shown in Figure 2A. The specificity for PLC versus healthy (100%; 95% CI, 96.8–100%) was higher than for PLC versus LC/HBV (96.1%; 95% CI, 86.5–99.5%) (Table 1). Boxplots using cancer scores predicted by the model demonstrated that the scores of patients with PLC were higher than those of healthy or LC/HBV controls as expected, whereas the LC/HBV group, despite its smaller group size, had more high-score outliers than the healthy controls (Figure 2B). The 10-fold cross-validation using the training cohort showed similar AUC differentiating PLC with different controls (vs. noncancer: 0.994; 95% CI, 0.988–1.000; vs. healthy: 0.999; 95% CI, 0.998–1.000; vs. LC/HBV: 0.983; 95% CI, 0.964–1.000), as shown in Figure S2.

We then constructed a predictive model using the raw coverage (4.45X–24.04X) WGS data, yielding an AUC of 0.994 (95% CI, 0.985–1.000) and a sensitivity of 95.2% (95% CI, 91.2%–97.8%) at 98.8% specificity (95% CI, 95.7%–99.9%) in differentiating patients with PLC from noncancer controls in the test cohort, as shown in Figure S3 and Table S3. Because the affordability of the WGS-based cancer detection method was primarily associated with low-coverage sequencing and the 4X coverage model slightly outperformed

the raw coverage model, we continued our evaluation using the 4X coverage model unless otherwise noted.

A heatmap analysis was performed to show the pattern for the FSD feature in both training and test cohorts. The top 100 features by variable importance were selected for heatmap construction. As shown in Figure 3A, there were noticeable differences between the patients with PLC and noncancer controls in the training cohort. A similar pattern for these features was observed within the test cohort (Figure 3B), further validating our model's power for differentiating PLC from noncancer controls. To further illustrate the advantages of using an ensemble stacked approach, we then compared our results against a logistic regression model using the same entire 4463 FSR and FSD features. As shown in Figure 2 and Figure S4, the overall AUC (PLC vs. noncancer) of the logistic regression model reached 0.988 (95% CI, 0.978–0.998), albeit showing excellent differentiating ability, was inferior to the ensemble stacked model (AUC: 0.995; 0.989–1.000). The AUC (0.965; 95% CI, 0.933–0.997) dropped slightly more while distinguishing patients with PLC from at-risk patients with LC/HBV using the logistic regression model. The high AUC by the logistic regression model further illustrated the excellent predicting power by the fragmentomics markers, which showed great differences among the patients with PLC and noncancer controls.

As shown in Figure S5A,B, the patients with PLC showed higher ratios of short fragments when compared with the noncancer control groups across the genome. Furthermore, the FSR feature profile and the copy number variation profile from 1 randomly selected patient with PLC showed a high correlation in regions with copy number gain/loss (Figure S5C,D), whereas the FSR features showed high alteration between the PLC and healthy groups even in the copy number neutral regions (Figure S5A,B). Our data agreed with previous findings that patients with cancer tended to have shorter cfDNA fragments compared with healthy controls.^[16]

Subgroup analysis and limit of detection

We then examined the model performances for different PLC and LC/HBV subgroups. As shown in Table 2, our model demonstrated exceptional capacity in detecting all PLC histology types. The sensitivity for detecting HCC reached 96.2% (95% CI, 91.9%–98.6%) and was even higher for detecting ICC (100.0%, 95% CI, 86.8%–100.0%) and cHCC-ICC (Table 2).

Our model showed excellent sensitivities when detecting patients with TNM early-stage HCC (stage IA: 90.0%; 95% CI, 68.3%–98.8%; stage IB: 96.6%; 95% CI, 90.3%–99.3%) in the test cohort (Figure 4A). As expected, the sensitivities for detecting patients with late-stage HCC were higher (stage II: 97.5%; 95% CI, 86.8%–99.9%; stage IIIA: 100.0%; 95% CI, 15.8%–100.0%; stage IIIB:

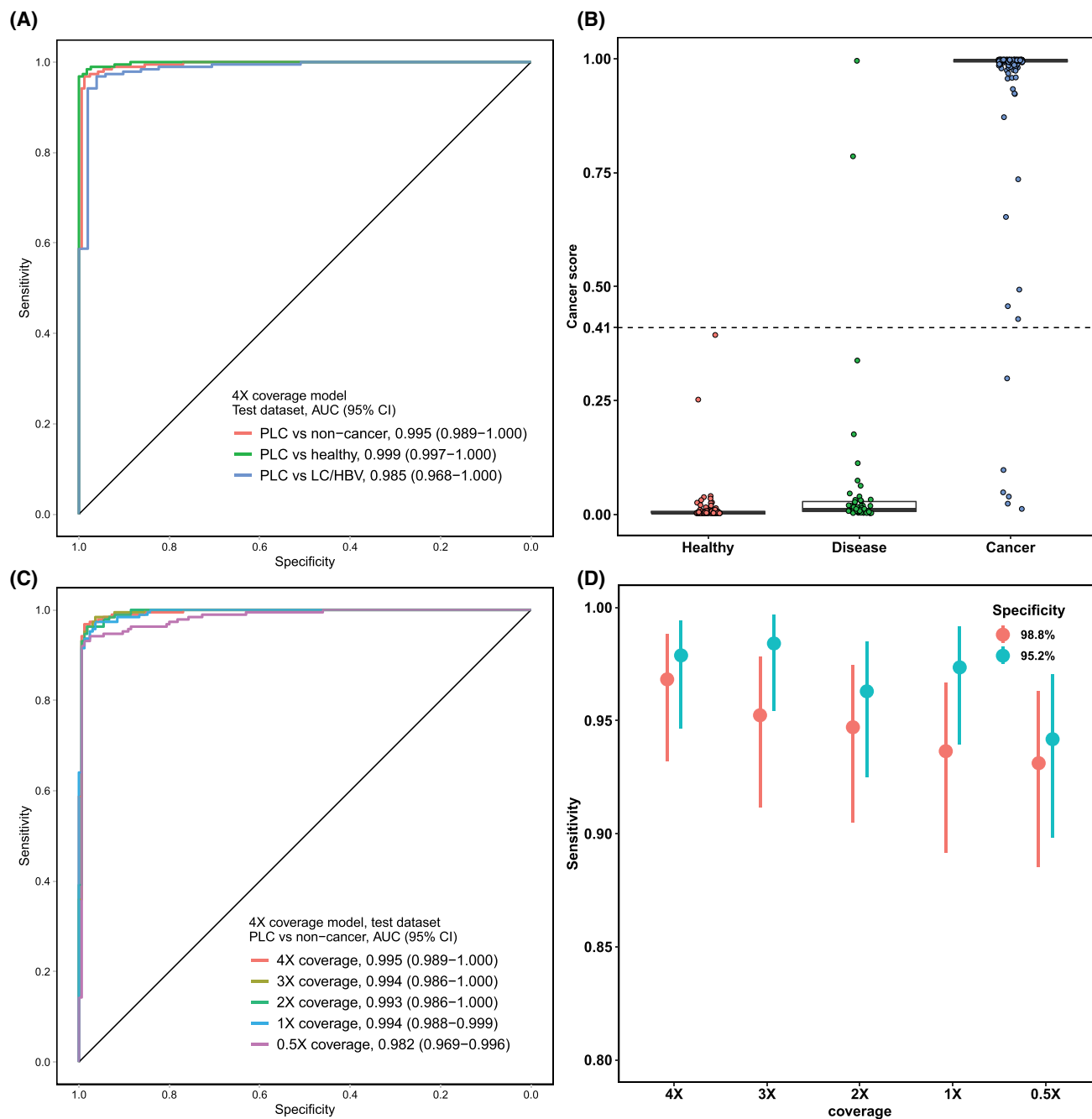


FIGURE 2 Evaluation of ensemble stacked model detecting early PLC. (A) ROC curves evaluating the overall performance of the predictive model, which was constructed using unified 4X coverage WGS data, in distinguishing patients with PLC from noncancer controls (LC/HBV, healthy) in the test cohort. (B) Boxplots illustrating cancer score distribution in the healthy, disease, and cancer groups in the test cohort based on the 4X coverage model. The 98.8% specificity cutoff for cancer score was 0.41, as shown by the dotted line. (C) ROC curves for distinguishing PLC from noncancer controls of a limit of detection analysis, the 4X coverage model was evaluated using WGS data down-sampled to 3X, 2X, 1X, and 0.5X. (D) Dot plot of 4X coverage model sensitivity in detecting PLC using 4X, 3X, 2X, 1X, and 0.5X WGS data at 98.8% (red) and 95.2% (green) specificities for noncancer controls. The error bars represented 95% CI

100.0%; 95% CI, 63.1%–100.0%), as shown in Table 2. A more detailed analysis, as shown in Table S4, suggested that our model performed even better in detecting TNM early stages in patients with ICC/cHCC-ICC (stage IA: 100%; 95% CI, 80.5%–100%; stage IB: 100%; 95% CI, 2.5%–100%) compared with patients with HCC (stage

IA: 90.0%; 95% CI, 68.3%–98.8%; stage IB: 96.6%; 95% CI, 90.3%–99.3%). This was likely contributed by the different criteria between ICC/cHCC-ICC and HCC in the AJCC TNM staging system, resulting in the inclusion of more advanced tumor under the same stages for ICC/cHCC-ICC.^[27]

TABLE 1 Evaluating model performances using the test dataset

Cancer vs. Noncancer		Actual	
		Cancer	Noncancer
Predict	Cancer	183	2
	Noncancer	6	163
Sensitivity (95% CI)		96.8% (93.2%–98.8%)	
Specificity (95% CI)		98.8% (95.7%–99.9%)	
PPV (95% CI)		98.9% (96.1%–99.9%)	
NPV (95% CI)		96.4% (92.4%–98.7%)	
Accuracy (95% CI)		97.7% (95.6%–99%)	
Cancer vs. healthy		Cancer	Healthy
Predict	Cancer	183	0
	Healthy	6	114
Sensitivity (95% CI)		96.8% (93.2%–98.8%)	
Specificity (95% CI)		100% (96.8%–100%)	
PPV (95% CI)		100% (98%–100%)	
NPV (95% CI)		95% (89.4%–98.1%)	
Accuracy (95% CI)		98% (95.7%–99.3%)	
Cancer vs. disease		Cancer	Disease
Predict	Cancer	183	2
	Disease	6	49
Sensitivity (95% CI)		96.8% (93.2%–98.8%)	
Specificity (95% CI)		96.1% (86.5%–99.5%)	
PPV (95% CI)		98.9% (96.1%–99.9%)	
NPV (95% CI)		89.1% (77.8%–95.9%)	
Accuracy (95% CI)		96.7% (93.5%–98.6%)	

Abbreviations: NPV, negative predictive value; PPV, positive predictive value.

The model performances were evaluated in different detailed subgroups, mainly to investigate whether specific subgroups were prone to errors. Our model was highly sensitive to detect the patients with very early-stage (BCLC stage 0/TNM stage IA) and early-stage (BCLC stage A/TNM stage IB) HCC. The sensitivities showed an upward trend from patients with BCLC stage 0/A (90%; 95% CI, 68.3%–98.8% and 96.7%; 95% CI, 91.7%–99.1%) to later-stage B/C (100%; 95% CI, 66.4%–100%; 100%; 95% CI, 63.1%–100%) (Figure 4A and Table 2). A similar pattern was observed for the CNLC staging, as our model showed a 94.7% sensitivity (95% CI, 88.8%–98%) for detecting stage Ia while achieving 100% sensitivities for later stages, as shown in Table 2.

In both the patient groups with PLC and LC/HBV, our model demonstrated a higher sensitivity/specificity (97.3%; 95% CI, 93.5%–99.1% vs. 83.3%; 95% CI, 35.9%–99.6% and 97.7%; 95% CI, 88%–99.9% vs. 80.0%; 95% CI, 28.4%–99.5%, respectively) for detecting total bilirubin level high subgroups (≥ 34 $\mu\text{mol/L}$ vs. < 34 $\mu\text{mol/L}$; Table 2). Similarly, the model showed great sensitivity/specificity within patient subgroups of PLC and LC/HBV with different alpha-fetoprotein (AFP) concentration levels (Figure 4 and Table 2). ROC curves were constructed distinguishing patients with PLC from patients with LC/HBV using both the AFP concentration and cfDNA fragmentomics model in the test cohort. As shown in Figure S6, our model showed better AUC (0.985; 95% CI, 0.968–1.000) compared with the traditional AFP biomarker (0.781; 95% CI, 0.715–0.848). As shown in Figure 4B and Table 2, specificity for detecting patients with LC/HBV increased (92.9%; 95%

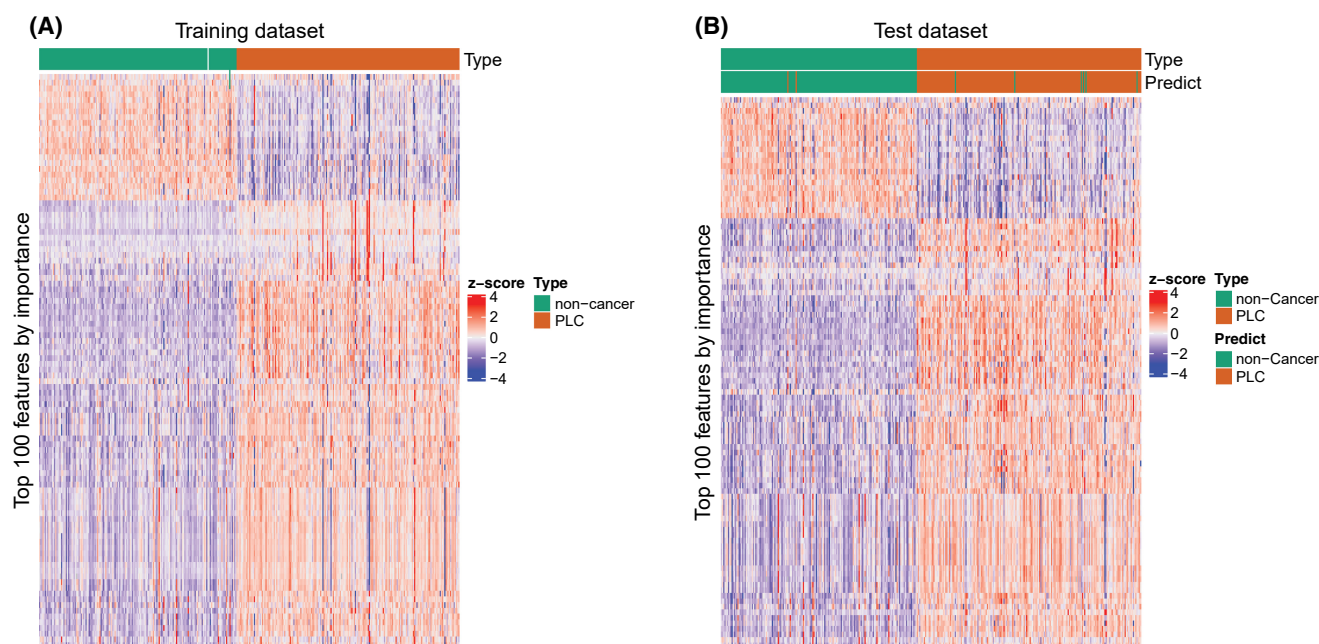
**FIGURE 3** Heatmap of fragmentomic features in the training and test cohorts. Heatmap of the top 100 features by variable importance in (A) training cohort and (B) test cohort

TABLE 2 Model performances for patient subgroups in the test dataset

Subgroups	Number	Sensitivity/Specificity	
PLC with PLC	Histology		
	HCC	157	96.2% (91.9%–98.6%)
	ICC	26	100.0% (86.8%–100.0%)
	cHCC-ICC	6	100.0% (54.1%–100.0%)
	Stage TNM for HCC		
	IA	20	90.0% (68.3%–98.8%)
	IB	87	96.6% (90.3%–99.3%)
	II	40	97.5% (86.8%–99.9%)
	IIIA	2	100.0% (15.8%–100.0%)
	IIIB	8	100.0% (63.1%–100.0%)
	Stage BCLC for HCC		
	0	20	90.0% (68.3%–98.8%)
	A	120	96.7% (91.7%–99.1%)
	B	9	100.0% (66.4%–100.0%)
	C	8	100.0% (63.1%–100.0%)
	Stage CNLC for HCC		
	Ia	114	94.7% (88.9%–98%)
	Ib	26	100.0% (86.8%–100%)
	Ila	7	100.0% (59.0%–100%)
	Ilb	2	100.0% (15.8%–100%)
	IIla	8	100.0% (63.1%–100%)
	Tumor size (cm)		
	<3	58	96.6% (88.1%–99.6%)
	3–5	100	96.0% (90.1%–98.9%)
	>5	31	100.0% (88.8%–100.0%)
	Tumor number		
	=1	162	96.3% (92.1%–98.6%)
	>1	27	100.0% (87.2%–100.0%)
	AFP (μg/L)		
	<20	114	99.1% (95.2%–100.0%)
	20–400	41	87.8% (73.8%–95.9%)
	>400	34	100.0% (89.7%–100.0%)
	TBIL (μmol/L)		
	<34	183	97.3% (93.7%–99.1%)
	>34	6	83.3% (35.9%–99.6%)
	ALB (g/L)		
	≤35	3	100.0% (29.2%–100.0%)
	>35	186	96.8% (93.1%–98.8%)
	LC history		
	No	81	98.8% (93.3%–100.0%)
	Yes	108	95.4% (89.5%–98.5%)

(Continues)

TABLE 2 (Continued)

Subgroups	Number	Sensitivity/Specificity	
Patients with LC/ HBV	AFP (μg/L)		
	<20	46	95.7% (85.2%–99.5%)
	20–400	3	100.0% (29.2%–100.0%)
	NA	1	100.0% (2.5%–100.0%)
	TBIL (μmol/L)		
	<34	44	97.7% (88.0%–99.9%)
	>34	5	80.0% (28.4%–99.5%)
	NA	1	100.0% (2.5%–100.0%)
	PLT (g/L)		
	<125	28	92.9% (76.5%–99.1%)
	125–350	19	100.0% (82.4%–100.0%)
	>350	2	100.0% (15.8%–100.0%)
	NA	1	100.0% (2.5%–100.0%)
	HBV		
	Neg	5	100.0% (47.8%–100.0%)
	Pos	45	97.8% (88.2%–99.9%)
	LC history		
	No	2	50.0% (1.26%–98.7%)
	Yes	48	97.9% (88.93%–99.9%)

Abbreviations: ALB, albumin; PLT, platelet; TBIL, total bilirubin level.

CI, 76.5%–99.1%; 100.0%; 95% CI, 82.4%–100.0%; 100.0%; 95% CI, 15.8%–100.0%) in accordance with their platelet level (<125 K/ml vs. 125–350 K/ml vs. >350 K/ml). Finally, patients' LC history and HBV status could slightly affect our model's differentiating power (Figure 4A,B), which was a possible limitation by their small numbers in the current cohorts. The large CIs for certain subgroups were likely contributed by the relatively small number of samples, which could be improved by enrolling more participants in future studies.

To assess the impact caused by imbalanced sex within our cohorts, we examined the differentiation power for both male and female individuals. As shown in Figure S7A, our model showed overall excellent performances for both male (AUC = 0.997; 95% CI, 0.993–1.000) and female (AUC = 0.988; 95% CI, 0.971–1.000) individuals. The boxplot illustrated no significant differences in cancer scores of male and female individuals between different subgroups (*t* test *p* value: 0.103, 0.327, and 0.098 for healthy, LC/HBV, and PLC, respectively) (Figure S7B). A propensity score matching analysis was performed further to eliminate the imbalance between the PLC and noncancer groups while evaluating our model's performance. A subset consisting of 120 PLC, 29 LC/HBV, and 62 healthy controls was selected from the test cohort with matching age and sex. As shown in Figure S7C, the age/sex-matched subset still showed excellent AUC (0.990; 95% CI, 0.978–1.000) distinguishing PLC from noncancer controls

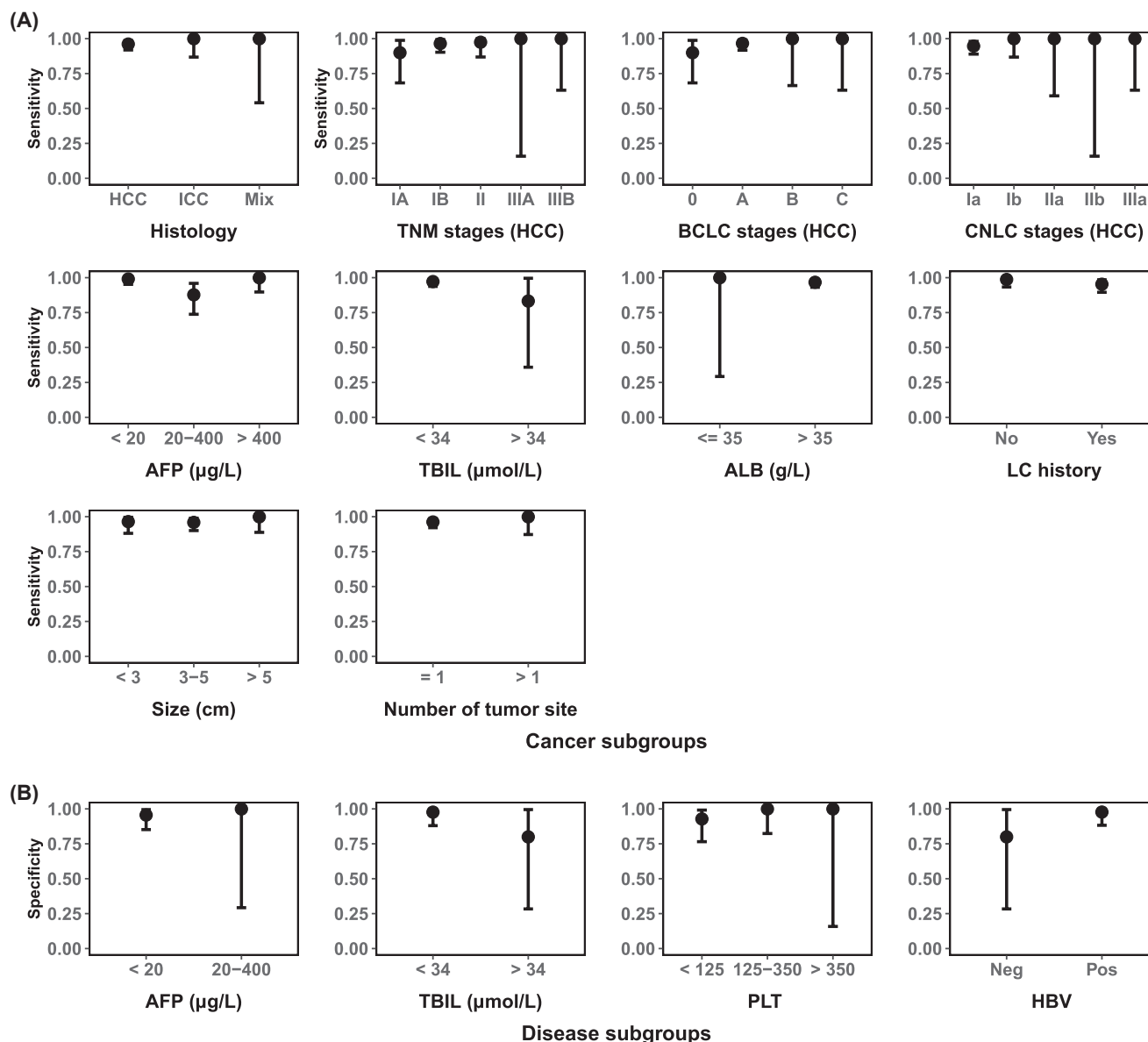


FIGURE 4 Evaluating sensitivity within (A) cancer and specificity within (B) disease subgroups. ALB, albumin; PLT, platelet; TBIL, total bilirubin level

compared with the entire test cohort (0.995; 95% CI, 0.989–0.997).

We then used even lower coverage WGS data to explore the limit of detection for our current model. The 4X coverage WGS data were further down-sampled to 3X, 2X, 1X, and 0.5X before being used for evaluating the 4X coverage model. As shown in Figure 2C, the differentiating power of our model was not affected by different WGS coverages judging by the AUCs (0.995, 0.994, 0.993, 0.994, and 0.982 for 4X, 3X, 2X, 1X, and 0.5X coverage data, respectively). However, as shown in Table S5, there was a gradual decrease in model sensitivity while using 3X (95.2%; 95% CI, 91.2%–97.8%), 2X (94.7%; 95% CI, 90.5%–97.4%), 1X (93.7%; 95% CI, 89.2%–96.7%), and 0.5X (93.1%; 95%

CI, 88.5%–96.3%) coverage data compared with 4X (96.8%; 95% CI, 93.2%–98.8%).

Furthermore, to investigate how the relatively small percentage of the at-risk patients with LC/HBV may have impacted our model, we constructed a serial of models with gradual removal of the patients with LC/HBV in the training cohort. These models were then tested using the same independent test cohort. As shown in Figure S8, whereas there was no visible performance decrease for the models' ability to distinguish the patients with PLC from the healthy controls, the mean AUC (5 repeats) for differentiating the patients with PLC and those with LC/HBV dropped to 0.974 while using only 20 patients with LC/HBV in the training cohort ($\sim 1/7$). However, the mean AUCs seemingly

reached a plateau using more than 45 patients with LC/HBV, suggesting that our model is suitable to be used for the screening PLC in an at-risk population.

Model differentiating HCC and ICC

Finally, we constructed an independent stacked ensemble model for distinguishing HCC and ICC using the same FSR and FSD profile. The model was trained using the 159 HCC and 26 ICC from the training cohort and evaluated using the 157 HCC and 26 ICC from the test cohort. As shown in Figure S9, the model illustrated that fragmentomic profiles had potential in distinguishing ICC from HCC, yielding an AUC of 0.776 (95% CI, 0.677–0.874). However, additional efforts are required to further improve the predictive power before potential clinical utilization.

DISCUSSION

This study, which enrolled participants with and without PLC, reported the development and validation of a fragmentomics-based machine learning model for PLC early detection using shallow WGS of plasma cfDNA samples.

In recent years, many studies have focused on the early detection of PLC because it is a leading cause of cancer mortality worldwide, with an estimated 5-year net survival of only 19%.^[2,3] However, most of the previous models reported so far have limited success in predicting PLC, with sensitivities of ~80–83% at the specificities of only ~76%–90%.^[12,13,17] Recently, Chen et al. reported a model integrating several cfDNA molecular features for HCC detection, achieving 95% sensitivity at the specificity of 97%.^[20] However, the HiFi model has used two sequencing approaches, genome-wide 5-hydroxymethylcytosine sequencing and low-coverage WGS, making it more complex, which in turn increases cost, both financially and time-wise. Our model, which relies solely on low-coverage WGS data, yielded an overall sensitivity of 96.8% for PLC detection at 98.8% specificity, and is therefore more suitable for clinical utilization. Our model maintained an excellent performance for PLC detection (93.7% sensitivity at 98.8% specificity) even using extremely low-coverage (1X) WGS data, which would help further reduce cost. Furthermore, when the sensitivity cutoff was set to 95.2%, there was no significant difference between sensitivities using the 4X (97.9%; 95% CI, 94.7%–99.4%) and 1X (97.4%; 95% CI, 93.9%–99.1%) WGS data.

Our early detection model's high performance was achieved by the utilization of different features and an ensemble stacked machine learning approach. It was shown by Jiang et al. that the cfDNA fragment sizes

were shorter in the plasma samples of patients with cancer.^[16] However, the detailed mechanism causing cfDNA fragment shortening in cancer samples remained unclear. Recently, Cristiano et al. developed a fragmentation-based approach for early cancer detection.^[18] Instead of just using the original fragmentation profile reported by DELFI,^[18] which compares the ratio of short and long reads in arbitrarily assigned bins across the genome, we have adopted two features called FSR and FSD. The FSR examined extended regions compared with the original DELFI short/long fragments ratio, and the FSD focuses on cfDNA FSD ratio at chromosome arm level in a high resolution of 5 bp. These optimizations can, in theory, capture more differences between cancer and noncancer groups. Our results confirmed that the base models using the FSR feature (93.7% sensitivity at 98.8% specificity) performed better than the original DELFI fragmentation GBM model (87.8% sensitivity at 98.8% specificity), whereas the stacked model (96.8% sensitivity at 98.8% specificity) further outperformed the base models.

One major concern for all machine learning models is the overfitting of training data. To prevent overfitting, we employed a stacked ensemble machine learning approach, which was known to reduce generalization error and overfitting.^[26] Moreover, we used two validation strategies, 10-fold cross-validation on the training cohort, and validation by an independent test cohort untouched during the model construction to evaluate our model performances, as they were the industry's gold standards to help prevent overfitting.^[30] The observation that consistently high performances were achieved in both evaluation assays indicated that our model is appropriate and the potential issue of overfitting, if any, has been mitigated. Finally, other biomarkers, such as plasma DNA methylation, have also shown adequately high AUC (0.96) differentiating patients with HCC from controls,^[15] providing additional reassurance for avoiding potential overfitting with cfDNA feature-based HCC prediction models.

Our model has also included the detection of ICC, which has been previously overlooked by the early detection studies on PLC. Our model showed an excellent performance while differentiating the ICC histology subgroup from the noncancer controls, achieving a 100% sensitivity at 98.8% specificity. Furthermore, preliminary data suggest that it is feasible to distinguish HCC and ICC using plasma cfDNA fragmentomic profiling, which could have certain clinical interests.

One limitation of our study is the relatively small number of at-risk patients with LC/HBV in both the training ($n = 53$) and test ($n = 51$) cohorts. Our model still showed great distinguishing power (AUC: 0.985) between patients with PLC and those with LC/HBV despite the low number of at-risk patients. Furthermore, our approach was able to maintain similar predicting power using fewer at-risk patients with LC/HBV in the

training cohort. However, this low number of at-risk patients with LC/HBV could still potentially affect our model's performances, as FNs in the group of patients with LC/HBV would result in an underestimation of specificity. Future studies with a larger at-risk patient cohort will be needed to better evaluate the performances of our PLC early-detection assay.

We believe a scalable, fully automated WGS library construction and analysis pipeline would be beneficial for using our assay for detecting PLC in a large at-risk population. Automation would help prevent artifacts introduced by possible human errors during library construction and therefore improve model performances. Furthermore, our automated pipeline costs less than 120 h per sample from collection to final prediction and is time-efficient for screening a large at-risk population.

In summary, we herein reported a predictive model using the comprehensive fragmentomic profiling of plasma cfDNA for PLC early detection. Our method, which is fast and affordable by relying on only low-coverage WGS data, has exhibited great potential in clinical practice for early detection of PLC and its different subtypes.

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CONFLICT OF INTEREST

Wanxiangfu Tang, Rui Liu, Hua Bao, Xin Chen, Yulin Wei, Shuyu Wu, Hairong Bao, Xue Wu, and Yang Shao are employees of Nanjing Geneseeq Technology Inc., Nanjing, Jiangsu, China. The remaining authors have nothing to declare.

AUTHOR CONTRIBUTIONS

Xiangyu Zhang (Conceptualization: Supporting; Data curation: Lead; Writing – review & editing: Equal). Zheng Wang (Conceptualization: Supporting; Data curation: Equal; Writing – review & editing: Lead). Wanxiangfu Tang (Methodology: Lead; Visualization: Lead; Writing – original draft: Lead; Writing – review & editing: Equal). Xinyu Wang (Investigation: Lead; Data curation: Supporting; Writing – review & editing: Equal). Rui Liu (Software: Equal; Methodology: Supporting). Hua Bao (Methodology: Equal; Writing – review & editing: Equal). Xin Chen (Visualization: Support; Writing – review & editing: Equal). Yulin Wei (Software: Equal; Methodology: Supporting). Shuyu Wu (Software: Equal; Project administration: Equal). Hairong Bao (Project administration: Lead; Writing – review & editing: Supporting). Xue Wu (Writing – review & editing: Equal). Yang Shao (Conceptualization: Equal; Supervision: Equal). Jia

Fan (Conceptualization: Equal; Supervision: Equal). Jian Zhou (Conceptualization: Lead; Funding acquisition: Lead; Supervision: Lead).

PATIENT CONSENT AND ETHICAL APPROVAL

All study protocols were approved by the Ethics Committee of Zhongshan Hospital Fudan University (Approval No. B2019-121R) and in accordance with international standards of good clinical practice. Written informed consent was provided by all patients.

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REFERENCES

- Bray F, Ferlay J, Soerjomataram I, Siegel RL, Torre LA, Jemal A. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin*. 2018;68(6):394–424.
- Dasgupta P, Henshaw C, Youlden DR, Clark PJ, Aitken JF, Baade PD. Global trends in incidence rates of primary adult liver cancers: a systematic review and meta-analysis. *Front Oncol*. 2020;10:171.
- Petrick JL, McGlynn KA. The changing epidemiology of primary liver cancer. *Curr Epidemiol Rep*. 2019;6(2):104–11.
- Kim SJ, Lee JM, Han JK, Kim KH, Lee JY, Choi BI. Peripheral mass-forming cholangiocarcinoma in cirrhotic liver. *AJR Am J Roentgenol*. 2007;189(6):1428–34.
- Singal AG, Mittal S, Yerokun OA, Ahn C, Marrero JA, Yopp AC, et al. Hepatocellular carcinoma screening associated with early tumor detection and improved survival among patients with cirrhosis in the US. *Am J Med*. 2017;130(9):1099–106.e1.
- Singal AG, Pillai A, Tiro J. Early detection, curative treatment, and survival rates for hepatocellular carcinoma surveillance in patients with cirrhosis: a meta-analysis. *PLoS Med*. 2014;11(4):e1001624.
- Patel M, Shariff MI, Ladep NG, Thillainayagam AV, Thomas HC, Khan SA, et al. Hepatocellular carcinoma: diagnostics and screening. *J Eval Clin Pract*. 2012;18(2):335–42.
- Stroun M, Maurice P, Vasioukhin V, Lyautey J, Lederrey C, Lefort F, et al. The origin and mechanism of circulating DNA. *Ann N Y Acad Sci*. 2000;906(1):161–8.
- Sun K, Jiang P, Chan KCA, Wong J, Cheng YKY, Liang RHS, et al. Plasma DNA tissue mapping by genome-wide methylation sequencing for noninvasive prenatal, cancer, and transplantation assessments. *Proc Natl Acad Sci U S A*. 2015;112(40):E5503–12.
- Benesova L, Belsanova B, Suchanek S, Kopeckova M, Minarikova P, Lipska L, et al. Mutation-based detection and monitoring of cell-free tumor DNA in peripheral blood of cancer patients. *Anal Biochem*. 2013;433(2):227–34.
- Jiang P, Sun K, Tong YK, Cheng SH, Cheng THT, Heung MMS, et al. Preferred end coordinates and somatic variants as signatures of circulating tumor DNA associated with hepatocellular carcinoma. *Proc Natl Acad Sci U S A*. 2018;115(46):E10925–33.
- Xu RH, Wei W, Krawczyk M, Wang W, Luo H, Flagg K, et al. Circulating tumour DNA methylation markers for

- diagnosis and prognosis of hepatocellular carcinoma. *Nat Mater*. 2017;16(11):1155–61.
13. Cai J, Chen L, Zhang Z, Zhang X, Lu X, Liu W, et al. Genome-wide mapping of 5-hydroxymethylcytosines in circulating cell-free DNA as a non-invasive approach for early detection of hepatocellular carcinoma. *Gut*. 2019;68(12):2195–205.
 14. Chan KCA, Jiang P, Chan CWM, Sun K, Wong J, Hui EP, et al. Noninvasive detection of cancer-associated genome-wide hypomethylation and copy number aberrations by plasma DNA bisulfite sequencing. *Proc Natl Acad Sci U S A*. 2013;110(47):18761–8.
 15. Kisiel JB, Dukek BA, Kanipakam RVSR, Ghosh HM, Yab TC, Berger CK, et al. Hepatocellular carcinoma detection by plasma methylated DNA: discovery, phase I pilot, and phase II clinical validation. *Hepatology*. 2019;69:1180–92.
 16. Jiang P, Chan CWM, Chan KCA, Cheng SH, Wong J, Wong V-S, et al. Lengthening and shortening of plasma DNA in hepatocellular carcinoma patients. *Proc Natl Acad Sci U S A*. 2015;112(11):E1317–25.
 17. Jiang P, Sun K, Peng W, Cheng SH, Ni M, Yeung PC, et al. Plasma DNA end-motif profiling as a fragmentomic marker in cancer, pregnancy, and transplantation. *Cancer Discov*. 2020;10(5):664–73.
 18. Cristiano S, Leal A, Phallen J, Fiksel J, Adleff V, Bruhm DC, et al. Genome-wide cell-free DNA fragmentation in patients with cancer. *Nature*. 2019;570(7761):385–9.
 19. Mathios D, Johansen JS, Cristiano S, Medina JE, Phallen J, Larsen KR, et al. Detection and characterization of lung cancer using cell-free DNA fragmentomes. *Nat Commun*. 2021;12(1):5060.
 20. Chen L, Abou-Alfa GK, Zheng BO, Liu J-F, Bai J, Du L-T, et al. Genome-scale profiling of circulating cell-free DNA signatures for early detection of hepatocellular carcinoma in cirrhotic patients. *Cell Res*. 2021;31(5):589–92.
 21. Kohl M MKmisc: Miscellaneous functions from M. Kohl. 2021. Published 8 August 2021. [cited 2021 Oct 30]. Available from: <https://rdrr.io/cran/MKmisc/>
 22. Flahault A, Cadilhac M, Thomas G. Sample size calculation should be performed for design accuracy in diagnostic test studies. *J Clin Epidemiol*. 2005;58(8):859–62.
 23. Liu YZ, Chu R, Lee A, Gomersall CD, Zhang L, Gin T, et al. A surveillance method to identify patients with sepsis from electronic health records in Hong Kong: a single centre retrospective study. *BMC Infect Dis*. 2020;20(1):652.
 24. Bolger AM, Lohse M, Usadel B. Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics*. 2014;30(15):2114–20.
 25. Li H, Durbin R. Fast and accurate short read alignment with Burrows-Wheeler transform. *Bioinformatics*. 2009;25(14):1754–60.
 26. Zhang C, Ma Y. Ensemble machine learning: methods and applications. Springer; 2012.
 27. Amin MB, Greene FL, Edge SB, Compton CC, Gershengwald JE, Brookland RK, Meyer L, et al. The Eighth Edition AJCC Cancer Staging Manual: Continuing to build a bridge from a population-based to a more “personalized” approach to cancer staging. *CA Cancer J Clin*. 2017;67(2):93–9.
 28. Llovet JM, Brú C, Bruix J. Prognosis of hepatocellular carcinoma: the BCLC staging classification. *Semin Liver Dis*. 1999;19(03):329–38.
 29. Xie DY, Ren ZG, Zhou J, Fan J, Gao Q. 2019 Chinese clinical guidelines for the management of hepatocellular carcinoma: updates and insights. *Hepatobiliary Surg Nutr*. 2020;9(4):452–63.
 30. MacKay EJ, Stubna MD. Understanding basic concepts of supervised machine learning model development in the clinical setting. *J Cardiothorac Vasc Anesth*. 2021;35(8):2336–7.

SUPPORTING INFORMATION

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